EVERY DERIVATION OF THE RING ACTS CANONICALLY ON THE LOCAL COHOMOLOGY OF AN IDEAL AS AN ADDITIVE OPERATOR SATISFYING THE LEIBNIZ RULE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The Cech complex of a finitely generated ideal computing R\Gamma_Z, with Z = V(I). Recalled inside the counted proof at source lines 4190-4196 and attributed to a lemma of another chapter (a cross-chapter pointer). Gives an explicit complex on which the derivation can act degreewise.
- Unique extension of a derivation of the ring to any localisation, still satisfying the Leibniz rule. Used inside the counted proof at source lines 4198-4200. Produces the degreewise action on the localised terms of the complex.

Source reference. Original reference dualizing-lemma-local-cohomology-adjoint: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 335–335. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-local-cohomology-adjoint, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a ring. Let $I \subset A$ be a finitely generated ideal. Set $Z = V(I)$. For each derivation $\theta : A \to A$ there exists a canonical additive operator D on the local cohomology modules $H^i_Z(A)$ satisfying the Leibniz rule with respect to θ.*

Proof. Let $f_1, \ldots, f_r$ be elements generating I. Recall that $R\Gamma_Z(A)$ is computed by the complex

$$A \to \prod\nolimits_{i_0} A_{f_{i_0}} \to \prod\nolimits_{i_0 < i_1} A_{f_{i_0} f_{i_1}} \to \ldots \to A_{f_1 \ldots f_r}$$

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See Dualizing Complexes, Lemma dualizing-lemma-local-cohomology-adjoint. Since θ extends uniquely to an additive operator on any localization of A satisfying the Leibniz rule with respect to θ, the lemma is clear. $\square$